Topological Information Aggregation Network for Few-Shot Cross-Domain Hyperspectral Image Classification

Kai Shi, Wenzhen Wang, Qichao Liu, *Member, IEEE*, and Liang Xiao, *Senior Member, IEEE*

***Abstract*— In recent advancements, hyperspectral image (HSI) classification through few-shot learning (FSL) has significantly progressed. Domain adaptation, integrated with FSL, effectively utilizes transferable knowledge from a source domain (SD) with abundant labeled data to excel in classification tasks within a target domain (TD) with scarce labels. However, most existing methods usually use traditional convolutional neural networks (CNNs) to extract local spatial information to characterize and mine feature and distribution information while ignoring the underlying topological relationships among feature classes. Therefore, we propose a topology graph perception cross-domain FSL (TGP-CFSL) framework that leverages graph information aggregation. Specifically, to construct the extended topological relationships of the target, we have designed a topological graph-based multiscale fusion (TGMF) feature extraction module, which is adept at fully mining the topological spatial neighborhood information of the target. Meanwhile, a dual-graph information perception (DGIP) module is designed, which is able to characterize and aggregate intradomain topological relationships in terms of both feature representations and interdomain distribution similarities and to extract higher order domain distribution information for realizing domain alignment. Experimental results on three public HSI datasets demonstrate that the proposed method outperforms existing methods.**

***Index Terms*— Cross-domain, few-shot learning (FSL), graph convolutional network (GCN), hyperspectral image (HSI) classification.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are 3-D data images containing rich spectral and spatial information formed by hundreds of consecutive narrow imaging bands captured by HSI in the wavelength range of 300–2500 nm. Since different substances have different absorption/reflection properties for electromagnetic waves of different wavelengths [1], the spectral curves presented can be used to identify the type of land cover at the pixel level. Due to this property, at present, HSI classification has been proved to be an indispensable key technology in the field of HSI remote sensing, and it has been widely applied in the fields of environmental monitoring [2], geological exploration [3], and agricultural assessment [4].

In recent years, with the continuous advancement of deep learning models, many machine learning-based [5], [6], [7], [8] and deep learning-based [9], [10], [11] approaches have been proposed to achieve single-domain HSI classification. Dedicated models for hyperspectral remote sensing data have made significant progress, especially with the introduction of the base model SpectralGPT [12]. SpectralGPT represents a pioneering effort in creating a universal spectral remote sensing foundation model, leveraging the generative pretrained transformer (GPT) architecture for the unique challenges of hyperspectral data. This model has demonstrated superior performance in various remote sensing tasks, including HSI classification, semantic segmentation, and change detection, due to its innovative 3-D masking strategy and multitarget reconstruction capabilities. However, sample labeling scarcity poses significant challenges in HSI classification due to resource limitations. Despite advances in unsupervised learning and semisupervised learning, these methods necessitate full model retraining with new data, incurring considerable overheads. In recent years, methods based on metalearning combined with domain adaptation have been proposed and gradually applied to HSI classification tasks, which extracts transferable knowledge from source domains (SDs) with abundant labeled samples, transfers models trained on SDs to target domains (TDs), and classifies them in conjunction with a small number of labeled samples in TD. However, there are still two main problems: 1) a large amount of distinguishable local feature information is ignored during the feature extraction process and 2) the domain shift problem is caused by the inconsistency in the distribution of the source and TDs, which seriously affects the classification effect.

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In order to solve the above problem, many methods have been proposed to eliminate the effect of domain bias. Ganin et al. [13] mitigated the effects of domain bias by introducing a domain classifier and a gradient inversion layer to explicitly encourage the network to learn domain invariant features. Ma et al. [14] learned to minimize the domain discrepancy by using an irrelevant task guide and maintained the domain alignment by sharing the parameters. Zhang et al. [15] further enriched the labeled database of TDs by introducing multiple sets of domain discriminators for domain alignment by extracting more domain information at different depths of the network and designing a pseudolabel refinement (PLR) module to select the most confident pseudolabeled samples for each class. Wang et al. [16] designed a cross-domain few-shot classification framework based on class attention that improves interclass decision boundaries by learning more discriminative features through a class attention metric module, enhances domain-invariant representations, and reduces the effect of domain bias through an asymmetric domain adversarial module. Although the above methods have achieved promising experimental results, they are all based on traditional convolutional neural networks (CNNs) to extract local spatial information and then characterize and align the data feature distributions in the source and TDs based on these features, and there are still two urgent problems to be solved due to their inherent characteristics.

A. Insufficient Spatial Information Representation

Traditional cross-domain few-shot HSI classification methods are usually based on CNN and its variants for feature extraction of hyperspectral data, on the basis of which the purposes of domain adaptation and few-shot learning (FSL) are accomplished. However, traditional convolutional methods are limited by local convolution and cannot construct effective remote topological relations for global targets, thus failing to mine more adequate spatial topological information, resulting in limited feature representation capability. Bai et al. [17] improved the network performance by introducing an attention mechanism and a feature transformation strategy, but this method still fails to fully mine more spatial information. Recently, GCNs have received much attention due to their ability to learn potential correlations between features. By propagating information along the spatial dimension, each feature target can obtain sufficient information from the entire HSI. For example, e.g., Qin et al. [18] proposed a semisupervised GCN to propagate messages between neighboring land covers based on their spatial distance and spectral similarity. To further improve the efficiency of message propagation, Mou et al. [19] proposed a nonlocal GCN by creating edges between neighboring and remote nodes, enabling long-distance land covers to exchange information directly without multiple convolutions. However, due to the nature of propagating features along the spatial dimension, GCN-based methods usually require the input of the entire HSI, leading to huge consumption of computational resources. To address this issue, researchers have begun using superpixels instead of pixels to construct graphs [11], [20], [21], [22]. Although this approach somewhat reduces the computational demands of GCNs, processing large-scale HSIs remains a formidable challenge.

B. Insufficient Modeling of Intersample Relationships

In recent years, graph networks have been increasingly used in cross-domain few-shot methods to mine data distribution relationships [23], [24], [25] because of their excellent ability to generalize and represent a wide variety of data structures and their distributions. Garcia and Bruna [26] proposed to construct a complete graph network in a few-shot task, where each node feature is concatenated with its corresponding class label and then through the graph network's attention mechanism to update the node features to aggregate distribution information. To further exploit the intracluster similarity and intercluster dissimilarity in graph-based networks, Kim et al. [27] proposed an edge-labeling graph neural network (EGNN) demonstrating an edge-labeled graph neural network in a contextual training framework. In addition, Sun et al. [28] used an adaptive pairwise graph learning technique to refine domain-invariant features while adaptively optimizing instance- and feature-level graphs, by which the data distribution in each domain can be efficiently preserved to accurately capture the intrinsic local flow structure of the input data in the low-dimensional flow subspace. However, it is worth noting that most of the above graph-based metalearning methods explore feature distribution information by modeling instance-level relationships of samples, and the main research focuses on pairwise relationships such as node labeling or edge labeling, ignoring a large number of substantial distribution relationships, which may lead to poor performance in the recognition of new scenes or new categories due to the lack of an in-depth understanding of the relationships of complex data.

To solve the above problems, we propose a topology graph perception cross-domain FSL (TGP-CFSL) framework. First, in order to solve the problem of insufficient feature information representation, a topological graph-based multiscale fusion (TGMF) feature extraction module is designed to complete feature extraction, which constructs a composite neighborhood containing both local and nonlocal neighbors and explores the composite neighborhood information of the target by a joint convolution module combining CNN and GCN. In addition, a dual-graph information perception (DGIP) module is designed to characterize and aggregate the data distribution in the domain from both feature- and distribution-level aspects and to generate higher order domain distribution information for achieving domain alignment. The main contributions of this article are given as follows.

1) To fully utilize the topological structure information of HSIs, the target's medium-remotely constructed topological relationship is leveraged, and the spatial neighborhood information is mined. Subsequently, the instance-level feature map and the feature distribution relationship map are constructed based on the feature relationship. This process extracts the higher order domain distribution information and alleviates the domain offset problem.
2) In order to provide a larger range of topological spatial information on the feature target and improve the information utilization and feature fusion, we constructed the spatial topological relationship of the target from the viewpoint of a graph, on the basis of which we proposed

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a joint convolution module to extract the spectral–spatial features.

3) A DGIP approach is proposed, which explicitly utilizes global relations for modeling, mines feature distribution relations to better capture the similarity of interdomain distribution information, and mitigates cross-domain differences by averaging losses.

II. Related Work

A. Few-Shot Learning Based on Metrics

In this article, a conventional metalearning model is followed with an episodic learning method [29], [30], where each FSL task $\mathcal{T}$ is represented by a support set S and a query set Q, where S is the set of labeled samples and Q is the set of unlabeled samples used to evaluate the classifier. In constructing the metatask, the support set S contains C classes, where each class K labeled samples, and the query set Q contains different samples from the same classes as S. This constitutes a C-way K-shot task.

In this approach, metric-based strategies emphasize learning discriminative features that enhance the separability between classes in a new, learned metric space [29], [31], [32]. A typical metric learning model might use distance functions such as the Euclidean distance or more complex measures such as the learned Mahalanobis distances to compare feature representations

$$d(f_\theta(x_i),\, f_\theta(x_j)) \tag{1}$$

where f_θ represents a neural network transforming input samples x_i and x_j into a metric space and d denotes the distance metric. The model is trained episodically, mimicking the few-shot scenario during both the training and testing phases, enhancing its ability to generalize from few shots.

During the testing phase, each sample x_q in the query set Q is classified by computing its distance to all samples in the support set S, and the predicted class is the one corresponding to the minimum distance

$$p(x_q) = \arg\min_{x_i \in S} d(f_\theta(x_q),\, f_\theta(x_i)) \tag{2}$$

where x_q represents an unlabeled sample in the query set Q and p represents the category label predicted for x_q, which is based on the minimization of the distance between x_q and all samples in the support set S.

In the cross-domain scenario, taking the few-shot task of SD as an example, the FSL loss of its query samples is computed by the cross-entropy loss

$$\mathcal{L}_{\text{fsl}}^{s} = -\sum_{x_q \in Q} \sum_{c=1}^{C} y_c \log\left(\hat{p}\left(x_q\right)\right) \tag{3}$$

where C denotes the number of categories contained in the SD, y_c denotes the one-hot coded value corresponding to the true category label of x_q, and $\hat{p}(x_q)$ denotes the probability that the model predicts that the sample x_q belongs to category c. The calculation process for $\mathcal{L}_{\text{fsl}}^{t}$ is similar to that of earlier, but note that $\mathcal{L}_{\text{fsl}}^{t}$ is obtained based on the metatask constructed using training data labeled in TD after data augment.

B. Graph Convolutional Network

Graph convolutional networks (GCNs) are increasingly being used for their excellent ability to represent unstructured data, which is mainly represented by edges connected through neighboring nodes and edges between nodes with specific aggregation and updating rules. A standard layer of a GCN can be expressed by the following equation:

$$h_v^{(l+1)} = \sigma\left(\sum_{u \in \mathcal{N}(v)\cup\{v\}} \frac{1}{\sqrt{|\mathcal{N}(u)||\mathcal{N}(v)|}} \mathbf{W}^{(l)} h_u^{(l)}\right) \tag{4}$$

where $h_v^{(l+1)}$ denotes the new feature representation of node v at layer $l+1$, σ denotes the nonlinear activation function, and $\mathbf{W}$ denotes the learnable weight matrix for layer l, applied to the features of each node. This weight matrix allows the model to learn how to best incorporate information from its neighbors at each layer. $(1/(|\mathcal{N}(u)||\mathcal{N}(v)|)^{1/2}$ denotes the normalization factor, where $\mathcal{N}(u)$ denotes the set of neighboring nodes of node u, $\mathcal{N}(v)$ denotes the set of neighboring nodes of node v, and $|\cdot|$ denotes the number of elements in the set. This factor is used to regulate the contribution of neighboring node features to prevent unstable values or the influence of certain nodes from being overamplified.

In recent years, due to the strong correlation between HSI features and the unstructured distribution of targets, GCNs have been gradually used to mine the rich topological structure information between features [22], [33]. Wan et al. [34] developed a new dual interactive GCN that introduces dual GCN branches to capture spatial information at different scales and leverages the correlation of multiscale spatial information to refine the graph information. It is demonstrated that the refined graph information helps to enhance the representation of the model. Jia et al. [35] constructed an external graph based on the spatial adjacency between superpixels after segmenting the HSI into disjoint superpixels, extracted the hierarchical features, and integrated them into multiple scales, which improves the distinguishability of the model.

C. Domain Adaptation

Domain adaptation methods aim to solve the problem of inconsistency between the distribution of the TD data and the SD data, i.e., the domain bias problem. Traditional domain adaptation techniques can be classified into three categories, distribution-based alignment, feature selection-based learning, and subspace-based learning, and have been widely used to solve cross-domain HSI classification tasks with fewer labeled samples. Feature selection-based approaches [36] build on the existence of a shared subspace between two domains and construct a model by selecting common features between the source and TDs. Methods based on subspace learning [37] transform the two domains into subspaces with the same distribution, e.g., manifold alignment (MA) [38]. On the other hand, distribution alignment-based methods [39], [40] achieve domain alignment by shortening the distribution distance of data between different domains and are considered to be the most direct and effective methods. It mainly consists of two categories, i.e., difference-based methods and adversary-based methods. Adversary-based methods achieve domain adaptation through adversarial training, which

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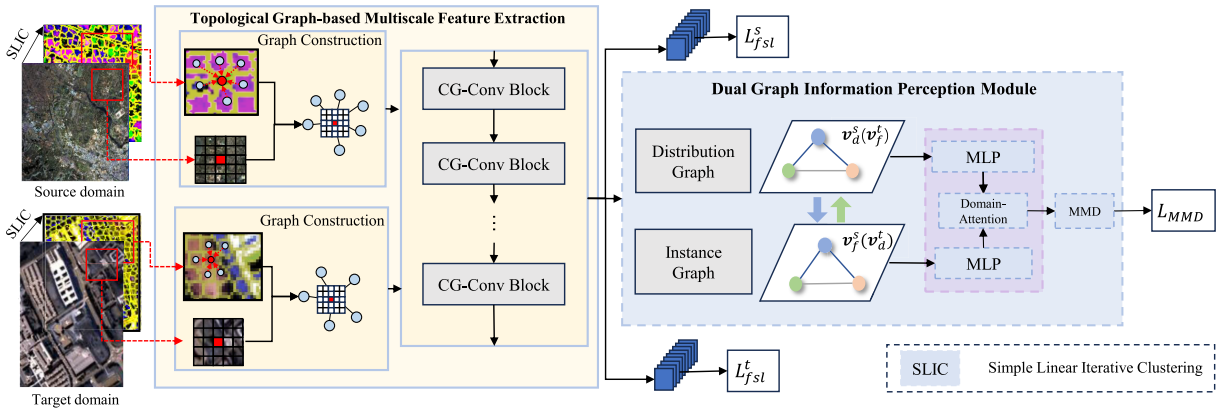


Fig. 1. Proposed TGP-CFSL framework. The network mainly consists of a feature extraction module based on composite convolution, a domain alignment module based on dual-graph information aggregation, and an FSL module. The TGMF module is utilized to extract the target features at multiple scales, and the DGIP module and FSL together guide the feature extraction module such that the extracted features are discriminative and domain invariant.

attempts to learn the domain-invariant feature space using two pairs of generators and discriminators in the source and TDs, where the discriminators replace the distance metric function that measures the difference in the difference-based distributions. Difference-based approaches optimize the model by introducing a distance loss function to achieve a consistent distribution of the generated features in both domains. However, most current distribution-based alignment methods do not focus on how the domains are aligned, i.e., they do not take into account the state of the data distribution within the aligned domains, unlike our method.

III. Proposed Method

The TGP-CFSL flowchart proposed in this article is shown in Fig. 1, which contains three components: feature extraction, FSL, and domain alignment. First, we construct composite neighborhoods in SD and TD, and in order to fully extract the features of the sample composite neighborhoods, we propose a joint convolution (CG-Conv) module for feature extraction. After obtaining the features of the target samples, we input the features into the DGIP module, and in the DGIP branch, we construct an instance graph (IG) and a distribution graph (DG) to enhance the categorical feature representation of the samples by propagating the class distribution information between the dual graphs. Then, we compute the similarity between SD and TD node distributions via the attention mechanism and compute the interdomain difference loss using the maximum mean difference (MMD). In addition, FSL uses the support and query sets constructed by composite neighborhoods to obtain the classification loss (cross-entropy loss) in a metalearning framework, which ultimately minimizes the weighted sum of the classification loss and the interdomain difference loss to jointly guide the network to update the parameters and optimize the network model.

A. Topological Graph-Based Multiscale Feature Extraction

In hyperspectral classification, traditional CNNs learn feature representations from a target's local square neighborhood, but they struggle to capture distant topological relations crucial for global analysis. To address this limitation, we integrate CNNs with GCNs to form a composite neighborhood, enhancing the sensory field. To be precise, we divide the HSI into spectrally similar and spatially connected hyperpixels; for each target pixel, a 9×9 square range around it is selected as its local neighborhood. Subsequently, the center nodes of several adjacent hyperpixels are then selected as the nonlocal neighbor nodes of the target pixel, and edges are established between the target local neighborhood and the nonlocal neighbor nodes to construct a graph. The local and nonlocal neighborhoods together constitute the composite neighborhood of the target pixel, and each sample consists of a target pixel and its composite neighborhood. We record the local neighborhood of a target and the nonlocal neighborhood on the constructed graph as $\mathbf{X} \in \mathbb{R}^{s\times s\times d}$ and $\mathbf{H} \in \mathbb{R}^{n\times d}$, respectively, where $s \times s$ denotes the spatial size of the local square region centered on that target, d is the size of the HSI spectral dimension, and n is the number of superpixel-centered nodes selected from around each target. The composite neighbor of the target can then be denoted as $(\mathbf{X}, \mathbf{H})$, which contains a total of $s \times s + n$ neighbor nodes. The support set $\{S_x, S_h\}$ and the query set $\{Q_x, Q_h\}$ are sampled from $\{\mathbf{X}, \mathbf{H}\}$.

In order to improve the utilization of composite neighborhood information and the efficiency of feature fusion, the designed CG-Conv is shown in Fig. 2. The proposed CG-Conv performs feature extraction by coupling CNN and GCN. For ease of description, we denote the input and output of the CG-Conv module as $(\mathbf{X}^{\text{in}}, \mathbf{H}^{\text{in}})$ and $(\mathbf{X}^{\text{out}}, \mathbf{H}^{\text{out}})$, respectively. For each pixel p_i, its composite neighborhood $N_i = (x_i \in \mathbb{R}^{s\times s\times d}, h_i \in \mathbb{R}^{n\times d})$, in each metatask $\mathcal{T}$, whether in the support set S or the query set Q, each sample is represented by the aforementioned N_i, that is, for the input of CG-Conv $(\mathbf{X}^{\text{in}}, \mathbf{H}^{\text{in}})$, where $\mathbf{X}^{\text{in}} \in \{S_x, Q_x\}$ and $\mathbf{H}^{\text{in}} \in \{S_h, Q_h\}$. The process of target feature extraction by conventional CNN convolution can be regarded as a local perception mechanism, in which the convolution operation is performed for each position of the input image, using the convolution kernel as a local window to perform a sliding operation in the local neighborhood. This process multiplies and sums the convolution kernel with the local neighborhood element by element, aggregating the information from the surrounding pixels to the central node. The basic paradigm is

$$\mathbf{X}^{\text{out}}_{i,j} = \sum_{m,n} \mathbf{K}_{(m,n)} \mathbf{X}^{\text{in}}_{i+m,j+n} \mathbf{W}. \tag{5}$$

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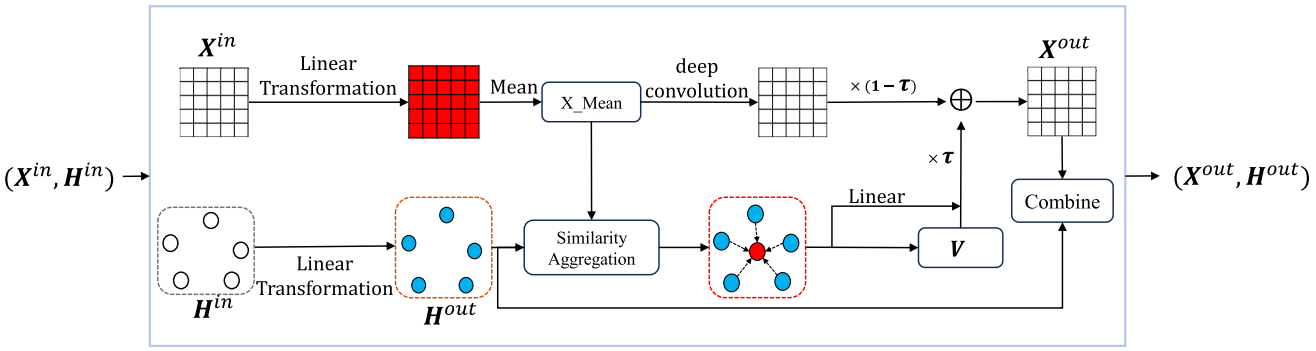


Fig. 2. Architecture of CG-Conv, where $(\mathbf{X}^{\text{in}}, \mathbf{H}^{\text{in}})$ and $(\mathbf{X}^{\text{out}}, \mathbf{H}^{\text{out}})$ denote the input and output data, and $\boldsymbol{\tau}$ is the fusion weights.

The bias term and the activation function are omitted here for ease of description. Here, $\mathbf{X}^{\text{in}}_{i+m,j+n}$ denotes the input pixel (row vector) at the location $(i+m, j+n)$, $\mathbf{K}_{(m,n)}$ is the learnable aggregation weight, and $\mathbf{W}$ is the learnable parameter of linear transformation. If we consider the aggregation weights as learnable parameters shared in the spatial dimension, the GCN can also be described as a process of neighbor node aggregation. The traditional general paradigm of GCNs is represented as

$$\mathbf{H}_i^{\text{out}} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \frac{1}{c_{ij}} \mathbf{W}\, \mathbf{H}_j^{\text{in}}\right) \tag{6}$$

where $\mathbf{W}$ denotes the learnable weight parameter of the lth layer, σ denotes the sigmoid function, and c_{ij} denotes the aggregation weight of node j to node i in the constructed graph. Unlike the fixed convolutional weights in CNN, the aggregation and updating of feature information of each node can be realized in graph convolution by using the learnable attention weights to weight and sum the features of neighboring nodes. Therefore, c_{ij} can also be obtained through the attention mechanism representation

$$c_{ij} = \frac{\sigma\left(\exp\left(e_{ij}\right)\right)}{\sum_{k\in\mathcal{N}(i)} \sigma\left(\exp(e_{ik})\right)} \tag{7}$$

where e_{ij} denotes the weights of the edges between node i and node j in the lth layer and σ denotes the sigmoid function. Therefore, based on the idea of aggregation, we try to unite the two kinds of convolution to aggregate the local neighborhood and nonlocal neighborhood information of the target nodes. Since the input local neighborhoods are 3-D vectors and the nonlocal neighborhoods based on the constructed graph representation are 1-D vectors, which makes it difficult to propagate features between them, the mean of the local neighborhoods of the target node is first computed first as the graph node

$$\mathbf{X}_{\text{mean}} = \frac{1}{s\times s}\sum_{i,j} \mathbf{X}_{i.j}. \tag{8}$$

Thus, the edge weight values in (7) are

$$e_{ij} = (\mathbf{W}\ \mathbf{X}_{\text{mean}}\ )(\mathbf{W}\ \mathbf{H}_i)^T. \tag{9}$$

Then, the nonlocal neighborhood information is aggregated to the target node using the attention mechanism, and according to (6) and (7), $\mathbf{H}_{\text{sim}}$ has been obtained

$$\mathbf{H}_{\text{sim}} = \sum_i \frac{1}{c_{ij}} \mathbf{H}_i. \tag{10}$$

At this point, all the information of the nonlocal neighborhood is aggregated in $\mathbf{H}_{\text{sim}}$. Finally, by combining the CNN convolution, we further fuse the local neighborhood information and nonlocal neighborhood information to obtain

$$\mathbf{X}_i^{\text{out}} = (\mathbf{1}-\boldsymbol{\tau}) \odot \sum_{m,n} \mathbf{K}_{(m,n)} \mathbf{X}^{\text{in}}_{i+m,j+n} \mathbf{W} + \boldsymbol{\tau} \odot \mathbf{H}^{\text{in}}_{\text{sim}} \tag{11}$$

where $\mathbf{1}$ denotes a vector with values all of 1, the operator $\odot$ denotes the elemental inner product, and $\boldsymbol{\tau}$ is a learnable attention weight parameter. Up to this point, by coupling image convolution and graph convolution, CG-Conv enables local and nonlocal neighborhoods to interact and integrate during feature extraction, thus improving the utilization of composite neighborhood information and feature fusion efficiency.

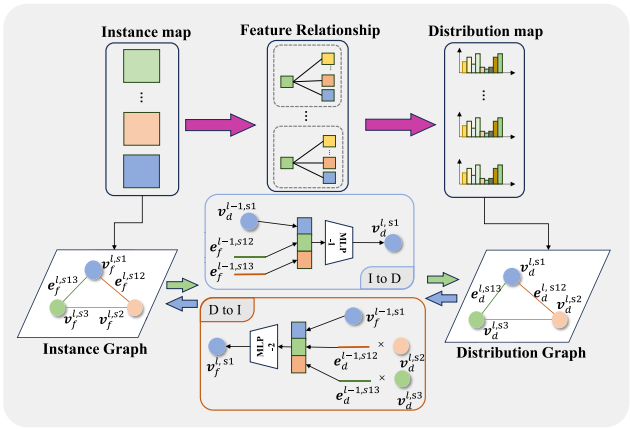


Fig. 3. Flowchart of the internal structure and computation of the dual graphs.

B. DGIP Module

In order to adequately represent the data distribution in the domain, we propose the DGIP module to extract the distribution information in the domain from both instance- and distribution-level aspects; first, we use the features extracted by the CG-Conv module to construct the instance-level graph $\mathcal{F} = (\mathbf{V}_f^{s/t}, \mathbf{E}_f^{s/t})$ and then use this to generate the distribution-level graph $\mathcal{D} = (\mathbf{V}_d^{s/t}, \mathbf{E}_d^{s/t})$. Taking SD as an example, the internal structure of the pairwise graph and the computational process are shown in Fig. 3.

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1) Node Initialization: The embedded features extracted by the feature extraction module are used as the initialization of the $\mathbf{V}_f^{s/t}$ node in the IG

$$\mathbf{v}_f^{s0} = F_{\text{CG-conv}}(\mathbf{X}, \mathbf{H}) \tag{12}$$

where $F_{\text{CG-conv}}(\cdot)$ denotes the joint convolution module of Section III-A, **(X,H)** denotes the composite neighborhood input, and $\mathbf{v}_f^s$ is a node in the IG denoting the feature class containing the target information. Correspondingly, the initialization of $\mathbf{E}_f^{s/t}$ edges in the IG is obtained by computing the similarity between neighboring nodes

$$\mathbf{e}_{f(i,j)}^{s0} = f_{\text{sim}}\left(\mathbf{v}_{f(i)}^{s0}, \mathbf{v}_{f(j)}^{s0}\right) \tag{13}$$

where $f_{\text{sim}}(\cdot,\cdot)$ denotes the similarity computation function, consisting of two Conv2d-BN-ReLU blocks, and $\mathbf{v}_{f(i)}^{s0}$ and $\mathbf{v}_{f(j)}^{s0}$ denote the initialized nodes in (9). For the DG, which is intended to be used to integrate instance relations from the IG and to process distributional relations, where each distributional feature $\mathbf{v}_d^{s0}$ is an NK-dimensional tensor, the value of the jth element denotes the relationship between the samples x_i and x_j, and NK denotes the total number of supported samples in the task. Therefore, for the initialization of each node

$$\mathbf{v}_d^{s0} = \begin{cases} \|_{j=1}^{\text{NK}} \delta\left(y_i, y_j\right), & \text{if } x_i \text{ is labeled} \\ \left[\dfrac{1}{\text{NK}}, \ldots, \dfrac{1}{\text{NK}}\right], & \text{otherwise} \end{cases} \tag{14}$$

where $\|$ and $[\cdot,\cdot]$ denotes the concatenation symbol and $\delta(\cdot,\cdot)$ denotes the Kronecker function, which outputs 1 if $y_i = y_j$ and 0 otherwise. The update process of the IG edge $\mathbf{e}_{d(i,j)}^{s0}$ is also similar to that of IG

$$\mathbf{e}_{d(i,j)}^{s0} = f_{\text{sim}}\left(\mathbf{v}_{d(i)}^{s0}, \mathbf{v}_{d(j)}^{s0}\right) \tag{15}$$

where $f_{\text{sim}}(\cdot,\cdot)$ denotes the similarity computation function, consisting of two Conv2d-BN-ReLU blocks, and $\mathbf{v}_{d(i)}^{s0}$ and $\mathbf{v}_{d(j)}^{s0}$ denote the initialized nodes in (10).

2) Dual-Graph Information Aggregation: We achieve information spreading and aggregation by building a bridge between two dual graphs in order to mine the distribution relationship of features in the domain and fine-tune the target feature information. Therefore, after the initialization of IG and DG, the information propagation aggregation process of the dyadic graphs is divided into two parts: one is the process of generating DG by IG, called I to D, and the other is the process of refining IG by DG by conveying the distributional relationship between each sample, called D to I. This cyclic transformation fully integrates the relationship between instance features and feature distributions.

a) Feature map to distribution map: The purpose of DG is to integrate instance-level relationships from IG to form distributional relationships and capture distributional information within the domain. Therefore, the similarity between IG nodes is used to aggregate the node characteristics for DG

$$\mathbf{v}_{d(i)}^{s1} = f_{\text{I2D}}\left(\|_{j=1}^{NK}\left(\mathbf{e}_{f(i,j)}^{s0}, \mathbf{v}_{d(i)}^{s0}\right)\right) \tag{16}$$

where $\|$ denotes the cascade notation and f_{I2D} denotes the aggregation network, which consists of a fully connected layer and ReLU.

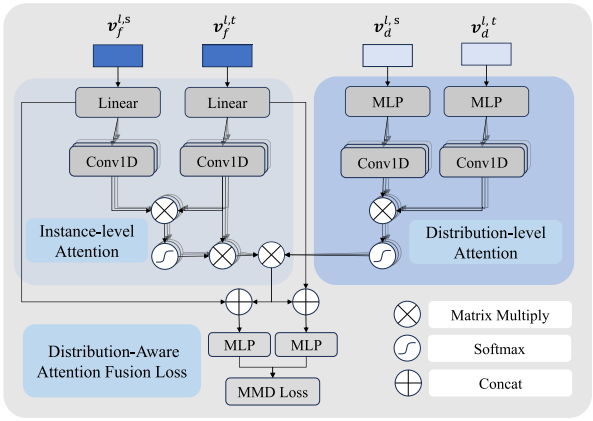


Fig. 4. Interaction of cross-domain information is achieved by generating higher order attention to the intradomain data distributions of SD and TD, learning the interdomain data distribution differences, and aligning the data distributions using the MMD to mitigate the domain bias problem.

b) Distribution map to feature map: Here the node features of the next layer of the IG are updated by the features of the edges of the DG and the node features of the IG

$$\mathbf{v}_{f(i)}^{s1} = f_{\text{D2I}}\left(\sum_{j=1}^{T}\left(\mathbf{e}_{d(i,j)}^{s1} \cdot \mathbf{v}_{f(j)}^{s0}\right), \mathbf{v}_{f(i)}^{s0}\right) \tag{17}$$

where f_{D2I} denotes the aggregation network, consisting of two layers of Conv2d-BN-ReLU, and T denotes the set of all samples participating in a single scenario-based training task.

C. Cross-Domain Few-Shot Learning

Then, the interaction of cross-domain information is realized by generating higher order attention to the intradomain data distribution of SD and TD, capturing the interdomain distribution similarity, learning the interdomain data distribution differences, and, finally, aligning the higher order attention to the distribution information using the MMD to alleviate the problem of domain bias. The process is shown in Fig. 4.

In order to accurately capture the data distribution differences between the fully captured domains, we mine the higher order data distribution features of SD and TD from both instance features and feature distribution aspects, respectively. Taking SD as an example, in the feature-level attention graph generation phase, the feature node $\mathbf{v}_{f(i)}^{s1}$ of the generated SD is used as a query, and $\mathbf{v}_{f(i)}^{t1}$ of the TD is used as key and value to obtain the feature-level attention representation of SD over TD

$$\alpha_{f(i,j)}^{s\to t} = \sum_j \sigma\left(\mathbf{W}_{\boldsymbol{\theta}}\mathbf{v}_{f(i)}^{s1}\left(\mathbf{W}_{\boldsymbol{\theta}}\mathbf{v}_{f(j)}^{t1}\right)^T\right)\mathbf{W}_{\boldsymbol{\theta}}\mathbf{v}_{f(j)}^{t1} \tag{18}$$

where σ denotes the softmax function and $\mathbf{W}_{\boldsymbol{\theta}}$ is the learnable weight. In order to obtain a wider range of interdomain distribution information, in the experiments, we used a multihead attention mechanism (four heads) to improve the expressive power of the model. To further capture the target and SD feature similarity, a temporary vector is set up

$$\mathbf{V} = \sum_j \alpha_{f(i,j)}^{s\to t}\mathbf{v}_{f(j)}^{t1}. \tag{19}$$

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In order to further fusion the distributional information of the features, $\mathbf{V}$ is used as the key, and $\mathbf{v}_{d(i)}^{s1}$ and $\mathbf{v}_{d(i)}^{t1}$ are used as the query and value for the attention fusion. In order not to lose the information of SD data, the residual structure is introduced, and the final output of the attention information of SD is obtained

$$I_{\text{out}}^{s} = \mathbf{v}_{f}^{s} + \text{MLP}\left(\left[\mathbf{v}_{f}^{s}, \alpha_{f(i,j)}^{s\to t}\right]\right) \tag{20}$$

where $[\cdot,\cdot]$ denotes the concatenation symbol and MLP denotes the multilayer perceptron. The distribution attention generation process corresponding to TD corresponds to the above process. At this point, I_{out}^{s} and I_{out}^{t} learn the higher order data distribution information of SD and TD by attentional fusion, and finally, maximum mean discrepancy (MMD) is introduced as a loss function to align the data distributions of SD and TD

$$\mathcal{L}_{\text{MMD}}\left(I_{\text{out}}^{s}, I_{\text{out}}^{t}\right) = \left\| \frac{1}{N}\sum_{i=1}^{N}\phi\left(I_{\text{out}}^{s}\right) - \frac{1}{N}\sum_{i=1}^{N}\phi\left(I_{\text{out}}^{t}\right) \right\|_{H}^{2} \tag{21}$$

where $N = C \times K + T$. Up to this point, we have mitigated the domain bias problem by advancing the effective domain data distribution spaces of SD and TD ever closer to each other by attentively learning the data distributions of the two domains continuously.

D. Cross-Domain Training and Testing

1) Training: The proposed framework is dedicated to solving the problem of few-shot HSI classification in cross-domain scenarios while enhancing the feature representation capability of the network and the data distribution characteristics. Following the general paradigm of metalearning, the classification loss (cross-entropy loss) during the metalearning process is computed from the query set prediction probabilities and their pairwise labels used in the training phase of the network, and then, the MMD loss of the interdomain distribution differences is extracted using the DGIP. Finally, the network parameters are updated by minimizing the weighted sum of the two losses

$$\mathcal{L}_{\text{total}} = \lambda_1(\mathcal{L}_{fsl}^{s} + \mathcal{L}_{fsl}^{t}) + \lambda_2(\mathcal{L}_{\text{MMD}}) \tag{22}$$

where λ_1 and λ_2 are the regularization parameters. More specifically, the training procedure of TGP-CFSL is available in Algorithm 1.

2) Testing: In the testing phase, discriminative and domain-invariant features of the training and test data in the TD are extracted using the trained network, and then, the features extracted from the labeled few-shot data are fit to obtain a nearest-neighbor classifier that is used for labeled prediction of the unlabeled test data.

IV. Experiments

All experiments in this article were conducted on a desktop computer equipped with an Intel Core i7-9700K CPU, 3.6 GHz, 24-GB RAM, and an Nvidia GeForce RTX 3090. In addition, considering the general requirement in the FSL setup that the number of classes in the SD data should be larger than the number of classes in the TD data as much as possible, the Chikusei dataset, which has more feature classes, is selected as the source scenario, and the University of Pavia (UP), the Indian Pines (IP), and the Salinas are selected as the target scenarios to validate the effectiveness of Gia-CFSL. Each dataset is fully described in this section.

A. Experimental Dataset

1) Chikusei Dataset: The Chikusei dataset was collected in 2014 by the hyperspectral visible/near-infrared C-series (Hyperspec-VNIR-C) sensor from Chikusei, Japan. The dataset contains 128 bands ranging from 368 to 1018 nm with a resolution of 2.5 m. Chikusei has a spatial size of 2517 × 2335 pixels and contains 19 classes such as water, soil, and vegetation, and the detailed information of each class is shown in Table I.

2) UP Dataset: The UP dataset was acquired in 2003 by the reflectance optical spectral imaging system (ROSIS-03) at the UP Campus, Italy. The dataset has a spatial resolution of 1.3 m and contains 610 × 340 pixels in the wavelength range 430–860 nm, applying 103 spectral bands by removing 12 bands corrupted by noise; 42 776 samples in total, nine classes, and these pixels can be classified into a total of nine categories, with detailed information for each category, as shown in Table I.

3) IP Dataset: The IP dataset was acquired in 1992 by the airborne visible/infrared imaging spectrometer (AVIRIS) at IN, USA. The dataset has a spatial size of 145 × 145 pixels and a spatial resolution of 20 m. It contains 16 categories and 10 249 samples, with detailed information for each category, as shown in Table I, where BGTD is the acronym for Buildings-Grass-Trees-Drives. The wavelength range of the IP dataset is 400–2500 nm with 220 bands. Experiments were performed using the remaining spectral bands, excluding the 20 bands affected by noise.

4) Salinas Dataset: It was taken by the AVIRIS sensor and is an image of the Salinas Valley, CA, USA. It has a spatial resolution of 3.7 m. After eliminating the bands that cannot be reflected by water, the image is left with 204 bands. The image contains 512 × 217 pixels, of which 54 129 contain real labels. In addition, these pixels are grouped into a total of 16 categories, and the details of each category are shown in Table I, where CSGW is the acronym for corn senesced green weeds.

B. Experimental Setting

TGP-CFSL is implemented in the PyTorch framework. The feature extraction module consists of $N = 3$ CG-Conv blocks. The composite neighbors of each target contain 9×9 pixel-level neighbors and 25 superpixel-level neighbors, i.e., $s = 9$ and $n = 25$. The average size of the superpixels in the simple linear iterative clusters (SLICs) is set to 64. In the training phase, the TGP-CFSL is trained using the standard Adam optimizer with a learning rate of 5×10^{-4}, a batch size of 64, and a number of training iterations of ten rounds of 10 000 iterations. In addition, for each metatask in the few-shot situational training C-way K-shot, C is set to the number of classes in TD, e.g., $C_{\text{UP}} = 9$ in the UP dataset, K is considered to be the random selection of K

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Algorithm 1 Training Procedure of TGP-CFSL

Input: HSI cube data: $X \in R^{9\times9\times Bands}$; Superpixel center node: $H \in R^{25\times Bands}$, Class number: N_c; Sample number per class in S and Q: N_s and N_q.

Output: The well-trained feature extractor F_θ.

for each $t \in [1, Iterations]$ **do**
  Randomly sample $T_t = S_t \cup Q_t$, $S_t = \{x_i, y_i\}_{i=1}^{N_c\times N_s}$, $Q_t = \{x_i, y_i\}_{i=N_c\times N_s+1}^{N_c\times(N_s+N_q)}$.
  // Topological Graph-based Multiscale Feature Extraction
  Compute the embedding data $X_{emb} \in R^{9\times9\times100}$, $H_{emb} \in R^{25\times100}$.
  Compute the features f_θ^j using CG-Conv.
  // Dual Graph Information Perception Module
  Initialize nodes v_f^0, v_d^0 and edges e_f^0, e_d^0 of feature graph and distribution graph using f_θ^j.
  Dual graph information aggregation.
  // Cross-domain Few Shot Learning
  Computing higher-order attention I_{out}^s by Eq.20 and $\mathcal{L}_{\text{MMD}}$ by Eq.21.
  Predict the class label of Q and compute the FSL loss $L_{fsl}^s(\theta)$ and $L_{fsl}^t(\theta)$;
  Update the parameters in CG-Conv and DGIP by minimizing $\mathcal{L}_{\text{total}}$ in Eq. 22.
end for

TABLE I
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN SOURCE AND TARGET DATASETS

Class No. \ Datasets	Chikusei		UP		IP		Salinas	
	Class Name	Numbers	Class Name	Numbers	Class Name	Numbers	Class Name	Numbers
1	Water	2845	Asphalt	6633	Alfalfa	46	Brocoli green weeds 1	2009
2	Bare soil (school)	2859	Meadows	18649	Corn-notill	1428	Brocoli green weeds 2	3726
3	Bare soil (park)	286	Gravel	2099	Corn-nimtill	830	Fallow	1976
4	Bare soil (farmland)	48525	Trees	30664	Corn	237	Fallow rough plow	1394
5	Natural plants	4297	Sheets	1345	Grass-pasture	483	Fallow smooth	2678
6	Weeds in farmland	1108	Bare soil	5029	Grass-tree	730	Stubble	3959
7	Forest	20516	Bitumen	1330	Grass-pasture-mowed	20	Celery	3579
8	Grass	6515	Bricks	3682	Hay-windrowed	473	Grape untrained	11271
9	Rice field (grown)	13369	Shadow	947	Oats	20	Soil vinyard develop	6203
10	Rice field (first stage)	1268	-	-	Soybean-notill	972	CSGW	3278
11	Row crops	5961	-	-	Soybean-miiitill	2455	Lettuce romaine 4wk	1063
12	Plastic house	2193	-	-	Soybean-clean	593	Lettuce romaine 5wk	1927
13	Manmade (non-dark)	1220	-	-	Wheat	205	Lettuce romaine 6wk	916
14	Manmade (dark)	7664	-	-	Woods	1265	Lettuce romaine 7wk	1070
15	Manmade (blue)	431	-	-	BGTD	336	Vinyard untrained	7268
16	Manmade (red)	222	-	-	Stone-Steel-Towers	93	Vinyard vertical trellis	1807
17	Manmade grass	1040	-	-	-	-	-	-
18	Asphalt	801	-	-	-	-	-	-
19	Paved ground	145	-	-	-	-	-	-
Total	-	77592	-	42776	-	10249	-	54129

labeled samples in each class to form the support set of T_s and T_t^l, and for a fair comparison, K is set to 1, and the number of samples in each class in the query set is set to 19. This is the same setting as in DCFSL, CA-CFSL, and Gia-CFSL.

Several state-of-the-art domain adaptive and FSL-based classification algorithms are used as comparison algorithms in the experiments, including SVM, 3D-CNN [41], SSRN [42], DFSL [43], CMFSL [44], DCFSL [39], SpectralFormer [9], CA-CFSL [16], and Gia-CFSL [45]. All of the above methods were experimentally accomplished to reproduce the comparison with the proposed methods. The evaluation parameters' overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa) are used to evaluate the classification performance of all models. Tables II–IV report the classification performance of the comparison algorithms on three test datasets, where five labeled samples are randomly selected from the TD as the training few-shot data. In addition, in order to verify the robustness of the experiments, the average metrics after ten runs are all used as the final results of the model tests.

C. Comparison With State-of the-Art Methods

In order to compare the classification performance of different methods more intuitively, Figs. 5–7 show the ground-truth maps and classification maps of the four test datasets, where the classification maps are reserved for the best results of the ten experiments. From the classification maps, the noise of the classification results based on the FSL method is overall less than that based on the non-FSL method and is closer to the ground truth, e.g., the no. 2 (meadows) and the class no. 6 (bare soil) of UP, the class no. 6 (grass-tree) and the class no. 11 (Soybean-miiitill) of IP, and the class no. 8 (grape untrained) and class no. 15 (Vinyard untrained) of Salinas.

To evaluate the performance of TGP-CFSL when SD and TD come from different sensors and contain different categories, we use related algorithms, such as SVM, 3D-CNN [41], DCFSL [39], SpectralFormer [9], Gia-CFSL [45], and CA-CFSL [16] for comparison.

Among them, SVM, 3D-CNN, and SpectralFormer methods belong to supervised non-FSL methods. A support vector machine is a traditional machine learning classification algorithm based on the principle of structural risk mini-

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TABLE II
CLASSIFICATION RESULTS (%) ON THE IP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3D-CNN [41]	SSRN [42]	DFSL [43]	CMFSL [44]	SpectralFormer [9]	CA-CFSL [16]	DCFSL [39]	Gia-CFSL [45]	TGP-CFSL
1	70.73±15.31	76.83±15.59	98.78±1.64	96.83±7.24	97.80±5.82	60.98±20.52	95.37±4.81	89.76±6.43	90.00±9.84	**99.27±2.90**
2	37.24±6.96	40.89±12.81	**57.49±12.20**	44.97±12.06	46.56±12.82	25.23±6.44	37.42±6.50	31.62±6.09	42.77±10.55	52.97±11.76
3	42.99±6.10	44.79±8.37	62.61±11.20	49.73±7.11	57.75±8.99	19.16±6.64	55.75±9.44	31,71±7.17	46.53±10.46	**69.92±7.31**
4	47.63±10.42	44.66±12.81	86.55±11.21	81.34±14.92	84.27±11.20	27.20±8.57	75.91±13.04	51.94±9.74	80.73±7.11	**91.16±10.70**
5	62.64±17.64	71.59±9.69	78.43±6.55	75.50±8.01	76.84±9.44	56.28±11.69	72.93±7.48	59.54±8.61	72.70±10.01	**80.92±9.55**
6	67.49±9.22	69.97±11.65	84.55±4.10	84.23±8.07	83.66±9.36	48.88±9.57	**87.45±5.84**	73.56±8.46	82.08±7.45	81.23±11.48
7	81.30±10.83	88.70±17.84	100.00±0.00	98.26±5.22	100.00±0.00	72.17±21.49	96.96±9.13	95.65±5.83	98.70±2.78	**100.00±0.00**
8	58.16±16.02	68.99±18.81	88.33±12.42	87.70±9.64	83.40±14.33	71.40±17.61	89.41±5.39	80.25±9.74	88.20±9.79	**97.82±4.55**
9	89.33±16.38	95.33±8.46	100.00±0.00	100.00±0.00	100.00±0.00	84.67±20.01	99.33±2.00	97.33±3.27	100.00±0.00	**100.00±0.00**
10	40.66±8.13	60.97±12.84	62.46±12.09	58.43±9.40	**64.92±7.33**	44.71±12.03	62.11±11.53	52.80±13.13	61.01±8.30	63.37±10.33
11	37.46±9.49	49.99±17.10	48.82±13.04	61.58±13.83	59.30±13.68	35.18±14.33	60.11±13.67	52.05±13.18	60.12±13.03	**72.35±5.87**
12	26.05±7.39	33.57±7.17	61.99±13.50	42.04±14.03	49.42±13.26	22.38±6.84	49.18±12.03	35.44±7.65	46.46±12.86	**71.07±6.82**
13	90.45±2.94	91.40±8.12	96.90±1.71	98.15±0.88	98.35±3.98	91.00±9.45	97.95±2.31	96.75±7.81	97.75±4.06	**98.40±2.12**
14	66.09±17.23	63.79±10.96	84.12±8.01	83.69±11.72	83.97±9.24	47.59±17.57	85.30±8.95	68.71±9.78	78.62±13.61	**86.25±8.57**
15	28.82±9.78	54.62±12.97	80.31±8.19	71.50±12.91	75.30±16.17	21.63±6.03	77.98±13.64	50.94±11.44	66.82±9.03	**92.94±10.30**
16	89.20±4.07	94.09±4.22	99.32±0.91	99.32±1.16	98.41±2.70	65.11±13.86	98.75±2.12	98.41±1.92	98.75±2.71	**99.43±2.37**
OA	47.07±2.20	54.93±4.53	68.30±2.84	65.25±2.98	66.68±2.01	38.77±3.04	66.83±2.06	54.28±3.23	63.70±3.65	**74.61±3.13**
AA	58.52±1.51	65.64±4.08	81.60±1.58	77.13±1.88	78.75±0.88	49.60±3.07	78.18±1.51	67.02±1.77	75.70±2.13	**84.82±1.56**
kappa	40.90±2.17	49.53±0.89	64.53±3.06	60.84±2.98	62.56±2.07	31.86±2.89	62.67±2.01	48.72±3.27	59.11±3.84	**71.47±3.41**

TABLE III
CLASSIFICATION RESULTS (%) ON THE UP DATASET (FIVE LABELED SAMPLES PER CLASS)

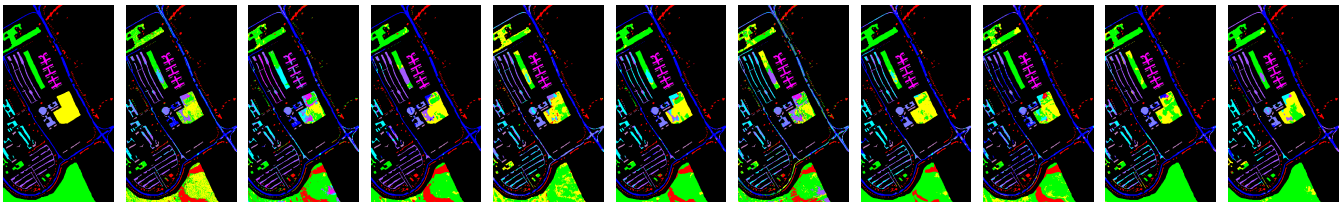
Class	SVM	3D-CNN [41]	SSRN [42]	DFSL [43]	CMFSL [44]	SpectralFormer [9]	CA-CFSL [16]	DCFSL [39]	Gia-CFSL [45]	TGP-CFSL
1	64.78±8.32	59.17±10.86	81.44±9.51	72.19±10.48	78.77±9.35	43.07±13.28	76.11±8.33	73.05±10.56	80.88±6.01	**82.30±7.98**
2	63.05±16.90	71.55±9.48	65.51±13.59	85.22±6.78	84.13± 6.57	65.34±8.08	75.92±9.68	78.00±8.43	**85.63±5.37**	83.89±6.16
3	48.91±12.29	65.03±12.33	68.49±15.40	60.09±13.15	66.02 ±10.70	49.28±11.65	52.48±12.73	48.98±10.86	60.65±8.93	**80.87±14.20**
4	80.90±11.42	74.67±6.30	91.79±6.25	92.63±2.48	93.67±3.46	72.23±7.89	91.37±4.16	87.54±6.26	**93.69±2.26**	88.36±8.60
5	98.68±1.05	86.16±8.15	99.06±1.81	98.72±1.07	99.66±0.22	91.14±12.21	98.76±1.05	98.42±1.60	99.36±1.06	**99.43±0.69**
6	54.06±12.97	39.35±7.02	**81.10±15.73**	69.88±12.12	76.71±12.09	42.55±8.77	59.47±8.65	57.71±8.51	75.65±9.22	74.39±11.66
7	69.11±17.68	83.42±12.11	87.92±6.80	73.06±9.30	79.52±8.93	77.37±6.65	70.05±7.69	71.36±9.12	77.17±8.60	**88.63±14.22**
8	68.86±13.15	65.88±12.23	81.39±11.19	58.01±14.12	79.23±10.76	46.76±14.13	60.36±9.03	63.91±13.77	72.36±10.90	**93.18±3.30**
9	**99.73±0.26**	92.44±6.54	99.63±0.81	98.22±1.32	99.18±1.13	95.72±3.12	98.25±1.83	98.34±2.08	96.01±5.60	90.62±11.05
OA	65.46±5.75	66.55±4.47	76.31±5.16	78.76±2.69	82.48±2.30	59.17±4.45	73.66±3.91	73.78±3.04	82.33±2.52	**84.27±2.68**
AA	72.01±2.15	70.85±3.07	85.04±3.18	78.78±2.55	84.11±1.56	64.83±2.70	75.86±1.19	75.26±1.67	82.38±1.65	**86.82±3.29**
kappa	56.70±5.73	57.72±4.71	70.53±5.62	72.45±3.19	77.39±2.64	48.93±4.78	66.30±4.37	66.29±3.29	77.11±2.98	**79.60±3.24**

TABLE IV
CLASSIFICATION RESULTS (%) ON THE SALINAS DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3D-CNN [41]	SSRN [42]	DFSL [43]	CMFSL [44]	SpectralFormer [9]	CA-CFSL [16]	DCFSL [39]	Gia-CFSL [45]	TGP-CFSL
1	98.19±0.97	97.92±2.96	99.23±0.35	95.46±2.05	99.37±0.84	78.98±15.34	**99.60±0.64**	98.43±1.95	99.31±0.85	98.07±2.76
2	94.77±6.71	97.44±4.41	97.69±0.63	95.89±4.11	99.56±0.72	86.04±8.66	**99.69±0.40**	98.66±2.49	98.89±1.55	98.39±1.58
3	82.14±11.87	87.77±7.54	91.05±9.06	91.93±1.52	92.34±8.39	78.03±14.75	**92.68±11.04**	92.23±8.59	88.20±9.32	90.64±6.54
4	98.25±2.01	98.94±0.64	98.61±0.51	99.24±0.25	99.28±0.94	98.28±1.48	**99.67±0.28**	99.32±1.04	99.00±1.28	98.02±2.60
5	**94.96±2.65**	93.21±1.68	85.87±15.75	93.75±0.41	93.63±3.35	87.83±4.47	89.84±4.45	92.00±2.95	88.11±4.60	93.81±2.37
6	98.81±0.71	98.45±1.70	99.36±0.82	98.37±0.53	99.32±0.39	97.98±2.07	99.44±0.61	99.10±0.90	99.45±1.81	**99.49±0.80**
7	99.28±0.43	98.29±1.22	**99.88±0.14**	95.84±3.06	99.56±0.39	97.12±1.65	98.95±2.27	98.30±1.38	98.69±1.51	98.85±1.54
8	54.28±10.26	59.52±12.13	74.57±15.44	46.65±17.87	72.32±11.97	48.48±10.44	76.13±12.20	65.76±13.67	72.68±14.76	**80.98±6.02**
9	97.98±0.89	96.74±2.70	98.99±0.59	99.27±0.58	99.45±0.92	93.47±3.01	**99.68±0.59**	99.64±0.24	98.22±2.23	99.32±1.00
10	76.80±6.82	72.18±14.84	**87.98±4.06**	83.62±0.79	86.21±7.81	53.39±14.94	80.68±13.51	79.61±12.92	80.67±11.40	84.71±10.65
11	87.80±4.49	91.25±4.66	98.26±1.07	94.12±0.71	98.32±1.39	83.27±7.01	97.99±1.66	95.20±7.88	96.21±3.04	**98.59±1.47**
12	93.77±7.03	**99.72±0.37**	98.64±1.81	96.64±1.38	99.31±1.17	94.29±7.43	98.82±1.21	99.19±1.49	98.35±2.94	99.39±0.91
13	95.85±4.36	94.77±2.49	99.30±0.74	98.85±0.05	99.29±0.48	90.31±16.64	98.69±0.60	**99.57±0.49**	98.36±1.73	96.54±3.70
14	90.93±2.57	93.18±4.41	97.81±1.75	96.62±2.82	**99.00±0.76**	90.27±7.07	97.76±1.96	98.27±1.78	96.83±3.23	96.39±4.21
15	52.33±9.63	66.47±11.43	73.35±9.88	69.53±5.48	77.40±8.97	58.05±12.43	74.10±9.07	63.48±11.55	75.32±7.38	**89.33**±5.37
16	82.97±7.83	82.67±7.22	90.48±6.66	91.31±6.24	91.76±5.74	72.06±13.26	90.03±5.13	**92.56±3.08**	88.84±6.71	91.32±6.74
OA	79.64±2.59	82.71±2.12	88.72±2.52	81.82±2.11	89.21±1.97	74.71±3.17	88.92±2.67	85.26±2.02	88.72±2.03	**92.12±1.96**
AA	87.45±1.64	89.28±1.40	93.46±1.50	90.67±0.36	94.15±0.96	81.74±4.57	93.35±1.51	91.98±1.48	93.22±0.86	**94.60±1.59**
kappa	77.43±2.85	80.85±2.31	87.47±2.75	79.90±2.25	88.02±2.16	72.02±3.53	87.69±2.93	83.63±2.19	87.18±2.83	**91.25±2.18**

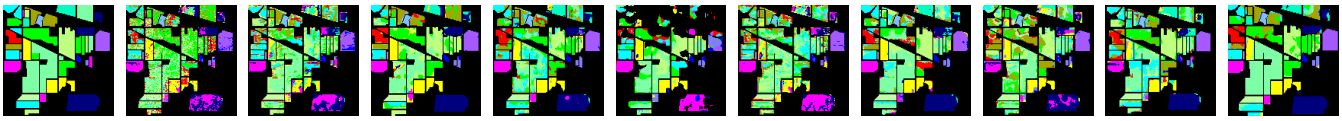
mization. With a robust theoretical foundation, it has been extensively applied and is notably effective in addressing the challenges of FSL in hyperspectral data within the computer science domain. However, SVM methods do not utilize the rich spectral information in the hyperspectral spectrum, while the 3D-CNN methods notice this and improve the classification accuracy by extracting the joint spatial–spectral features of the hyperspectral data. The SpectralFormer method rethinks the classification accuracy of the hyperspectral data from a Transformer's sequential perspective to rethink HSI classification, which is able to learn spectral local sequence information from neighboring bands of HSIs and generate grouped spectral embeddings, achieving good results without using any convolutional or cyclic units. In addition, in order

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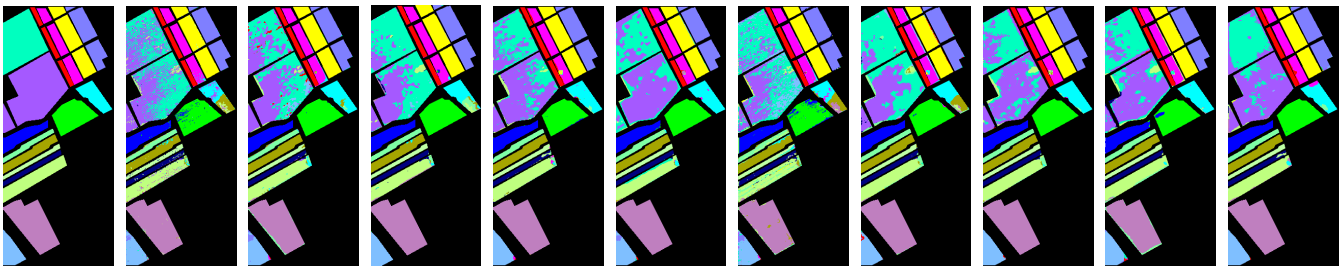
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 5. UP. (a) Ground truth. Classification maps for different classifiers. (b) SVM. (c) 3-D CNN. (d) SSRN. (e) DFSL. (f) CMFSL. (g) SpectralFormer. (h) CA-CFSL. (i) DCFSL. (j) Gia-CFSL. (k) TGP-CFSL.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 6. IP. (a) Ground truth. Classification maps for different classifiers. (b) SVM. (c) 3-D CNN. (d) SSRN. (e) DFSL. (f) CMFSL. (g) SpectralFormer. (h) CA-CFSL. (i) DCFSL. (j) Gia-CFSL. (k) TGP-CFSL.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 7. Salinas. (a) Ground truth. Classification maps for different classifiers. (b) SVM. (c) 3-D CNN. (d) SSRN. (e) DFSL. (f) CMFSL. (g) SpectralFormer. (h) CA-CFSL. (i) DCFSL. (j) Gia-CFSL. (k) TGP-CFSL.

to test the ability of SVM, 3D-CNN, and SpectralFormer methods to solve few-shot problems, five labeled TD samples are selected for training in each class, and the remaining samples are used as test samples.

DCFSL, Gia-CFSL, and CA-CFSL methods represent the latest supervised FSL methods. The DCFSL method adds a symmetric adversary-trained domain discrimination module to the network to minimize the effect of domain bias. The Gia-CFSL method combines the FSL with domain alignment based on graph information aggregation and focuses on nonlocal spatial information with strong correspondences during domain alignment. The CA-CFSL method introduces class attention to the framework of FSL to learn more discriminative features to improve interclass decision boundaries. The CA-CFSL method introduces class attention to the framework of FSL to learn more discriminative features to improve interclass decision boundaries. The CA-CFSL method introduces class attention to the framework of FSL to learn more discriminative features to improve the interclass decision boundary. In addition, we also introduced the idea of asymmetric domain adversarial to enhance the ability to extract domain invariant representations. For better comparison, DCFSL, Gia-CFSL, and CA-CFSL methods all use the Chikusei dataset as a common SD in their experiments. Among them, the DCFSL method draws 128 randomly labeled samples from the TD with few samples after data augmentation to learn the transferable features in the TD.

Tables II–IV show the comparison of the classification performance on the three target datasets, where five labeled samples are randomly selected from the TD for training. To test the robustness of the model, the experiment takes the average value after ten runs as the test result of the model, and the visualized classification results are shown in Figs. 5–7. The following conclusions can be drawn from the comparison results.

1) Among the supervised non-FSL (SVM, 3D-CNN, and SpectralFormer) methods, the classification accuracy of SpectralFormer, a Transformer-based method, is lower than that of SVM, a machine learning method, and methods based on deep learning network training on all the three datasets, and the analysis reveals that the Transformer-based models are susceptible to overfitting and lack of generalization in FSL scenarios due to their large number of parameters and their dependence on large datasets.
2) Compared with SVM, 3D-CNN, and SpectralFormer methods trained with only a few number of TD samples, DCFSL, Gia-CFSL, and CA-CFSL methods have 5%–11% higher classification accuracies on all three datasets, which is due to the latter's addition of different domain adaptation techniques, respectively, which improve the classification performance to a certain extent by minimizing the effect of domain bias.
3) Compared with the same type of HSI FSL cross-domain classification methods, the proposed TGP-CFSL method has better classification performance on the three datasets, and due to the introduction of intersample topology as an input to the network, CG-Conv can adaptively extract the useful information neighbors from the different levels without the interference of irrelevant,

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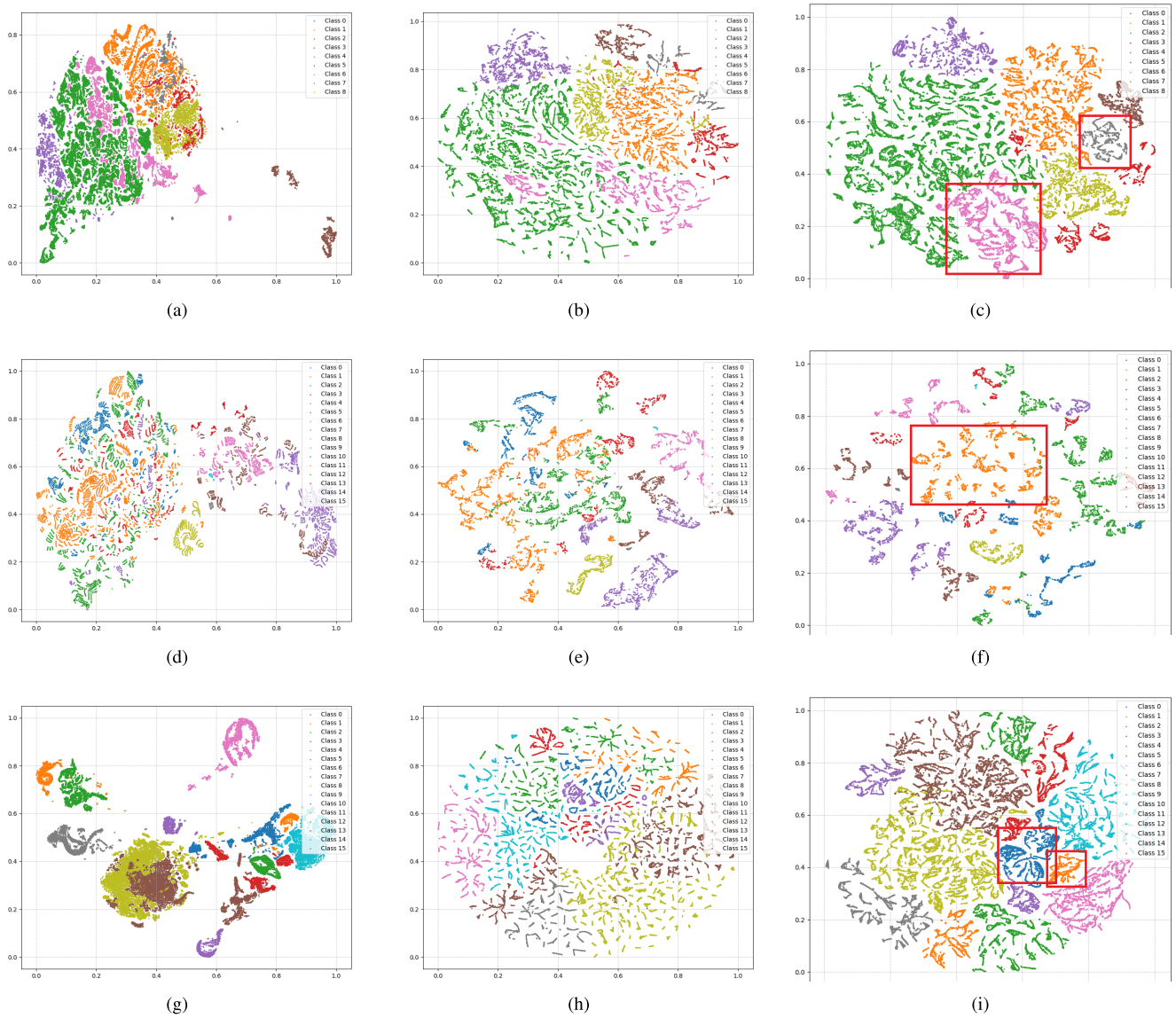


Fig. 8. Class separability of the proposed TGP-CFSL using Pavia University, IP, and Salinas, and it is obvious that features by TGP-CFSL have the best class separability. (a) Original samples from Pavia University. (b) Features by TGP-CFSL(w/o DGIP). (c) Features by TGP-CFSL. (d) Original samples from IP. (e) Features by TGP-CFSL(w/o DGIP). (f) Features by TGP-CFSL. (g) Original samples from Salinas. (h) Features by TGP-CFSL(w/o DGIP). (i) Features by TGP-CFSL.

which enables better classification of different land cover with a complex structure for modeling, thus producing excellent and stable classification results on various HSI datasets. In addition, graph structure information is further utilized in DGIP to align domain distributions both at the feature level and the distribution level, thus enabling a fuller perception of interdomain similarities to reduce cross-domain differences.

V. Discussion

To further investigate the TGP-CFSL, in this section, we conduct several experiments from various perspectives. In short, to prove the effectiveness of the proposed CG-Conv and DGIP module, a set of ablation experiments is performed. To analyze the influence of hyperparameters including the size of the learning rate and the number of training samples of each class in the support set, we set each hyperparameter to different values and then test the classification performance of the TGP-CFSL. Moreover, to further analyze the mechanism of the CG-Conv and DGIP modules, we conducted experiments on class separability and interclass prototype distance visualization on each of the three datasets. If not specified, all the experiments in this section are in the same configuration as in Section IV.

A. Ablation Study

The proposed TGP-DFSL framework consists of three components: the FSL, CG-Conv, and DGIP. In order to validate the proposed method and reflect the performance of each module of the proposed TGP-CFSL network, we conduct a series of ablation experiments. We constructed the baseline model by removing the nonlocal neighborhoods and DGIP modules in the composite neighborhood of each target, respectively, and replacing CG-Conv with a conventional 3D-

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TABLE V
ARCHITECTURE OF DIFFERENT MODULES

	Few-shot Learning	DGIP	CG-Conv
FSL	✓		
TGP-CFSL(w/o DGIP)	✓		✓
TGP-CFSL(w/o CG-Conv)	✓	✓	
TGP-CFSL	✓	✓	✓

TABLE VI
ABLATION COMPARISON OF EACH VARIANT OF TGP-CFSL (FIVE LABELED SAMPLES FROM TD)

Dataset	Model	FSL	TGP-CFSL (w/o CG-Conv)	TGP-CFSL (w/o DGIP)	TGP-CFSL
UP	OA	71.51±4.65	72.38±3.54	81.31±4.46	**84.27±2.68**
	AA	73.63±1.92	74.39±4.26	86.26±3.47	**86.82±3.29**
	kappa	63.59±5.04	64.20±3.92	76.05±5.34	**79.60±3.24**
IP	OA	54.28±2.75	59.18±2.53	74.06±3.57	**74.61±3.13**
	AA	67.58±1.04	72.46±1.58	84.67±2.04	**84.82±1.56**
	kappa	52.81±2.78	54.15±2.61	73.01±3.92	**71.46±3.41**
Salinas	OA	87.41±1.65	87.22±1.78	91.17±3.07	**92.12±1.96**
	AA	86.70±0.84	92.85±1.21	94.09±1.34	**94.60±1.59**
	kappa	87.14±1.80	85.83±1.94	90.95±1.66	**91.25±2.19**

CNN while preserving the basic network architecture of the FSL. Specifically, the ablation experiments shown in Table V are conducted separately.

1) FSL: It represents the baseline network with nonlocal neighbors in the composite neighborhood removed, CG-Conv degenerated into image convolution, and simple Euclidean distance used as a similarity metric.

2) TGP-CFSL(Without DGIP): It denotes that DGIP is removed from TGP-CFSL, the construction of the composite neighborhood is retained, and feature extraction is performed with CG-Conv.

3) TGP-CFSL(Without CG-Conv): It denotes that CG-Conv is removed from TGP-CFSL, feature extraction is performed with simple image convolution, and the domain offset problem is mitigated by DGIP combined with FSL.

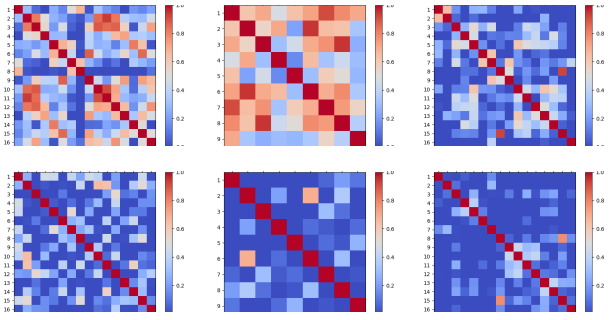
4) TGP-CFSL: It denotes the complete TGP-CFSL network.

Each experiment was repeated ten times to report OA, AA, and Kappa. As shown in Table VI, it is clear that the overall performance of TGP-CFSL is superior to the other variants and gains significant improvements. In particular, replacing the image convolution with CG-Conv brings a significant performance improvement. Compared to the baseline, the OA index of the network using CG-Conv increased by 9.80%, 10.86%, and 3.76% on the three datasets, respectively. Since the UP consists of a variety of land covers with a dispersed and complex spatial structure, nonlocal neighborhoods may contain less domain invariant feature information compared to local neighborhoods, resulting in a smaller performance improvement, whereas, on the Salinas dataset and the UP dataset, where most of the same land covers are clustered together, nonlocal neighborhoods can provide a large amount of information relevant to the classification, resulting in a significant performance improvement for this dataset.

In addition, the classification performance of "TGP-CFSL" is further improved by 2.96%, 0.55%, and 0.95% on the three datasets compared to "TGP-CFSL (without DGIP)", indicating that DGIP plays a role in domain alignment, which is fully characterized. It shows that DGIP plays a role in domain alignment by adequately characterizing the intradomain data distribution and aligning the domain distribution information to reduce cross-domain variations by capturing the similarity of interdomain distributions and learning the interdomain data distribution differences. In summary, although there are differences in the impact of each module on classification performance on different datasets, which is caused by the interdomain distance between different target datasets and the SD dataset, as well as the intradomain distribution differences of each dataset. Nevertheless, it is clear that all of the above modules have positive effects on improving the classification accuracy of the TGP-CFSL model.

TABLE VII
CLASSIFICATION RESULTS (%) OF THE PROPOSED NETWORK WITH DIFFERENT LEARNING RATES ON THE UP DATASET (FIVE LABELED SAMPLES PER CLASS)

Learning rate	0.05	0.005	0.0005
OA	71.59±4.86	79.38±3.91	**84.27±2.68**
AA	73.27±4.49	81.07±2.97	**86.82±3.29**
kappa	63.69±5.54	73.33±4.50	**79.59±3.24**



(a) (b) (c)

Fig. 9. 3D-CNN (top) and CG-Conv (bottom) visualize similarity maps between class-averaged feature representations across different datasets. Each map depicts the similarity values between prototypes of type i and type j, where row i and column j represent respective types. (a) IP. (b) UP. (c) Salinas.

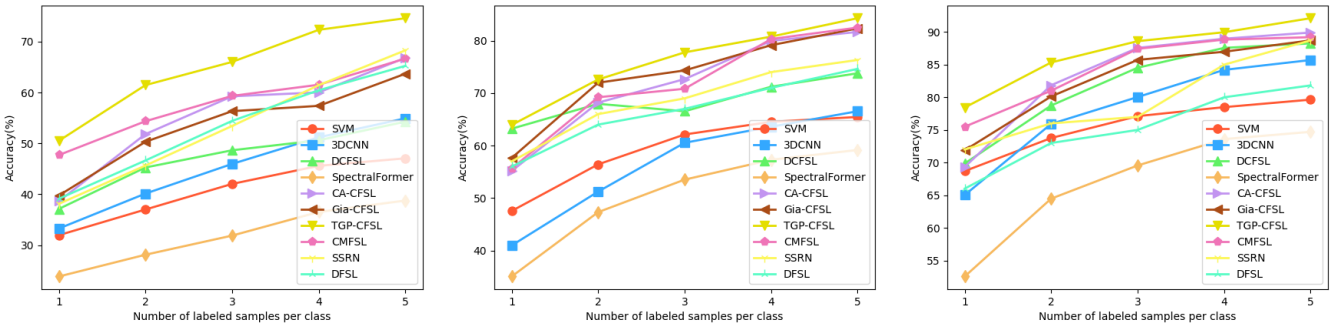
B. Alignment of Class Representation Features

To further demonstrate the usefulness of the TGMF and to analyze the results of the representation of the target features in the metric space, we visualize the similarity between classes represented by the mean of all target feature representations. As shown in Fig. 9, each similarity map, which can be regarded as a confusion matrix, presents the degree of network optimization in its current training epoch. Specifically, a larger similarity (red) between two prototypes indicates that their corresponding classes are more difficult to distinguish. On the contrary, a smaller similarity (blue) means that it is easier to classify their classes. The experimental results show that TGMF can improve the distinguishability of class features to different degrees. Among them, compared with traditional convolutional methods, TGMF can extract effective target features more adaptively according to the information of different neighboring targets and dynamically adjust the aggregation weights of neighboring nodes on the target features, so as to improve the distinguishability of feature representations.

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TABLE VIII
CLASSIFICATION RESULTS (%) ON UP DATASET UNDER DIFFERENT ω'S

ω		SVM	3D-CNN [41]	SSRN [42]	DFSL [43]	CMFSL [44]	SpectralFormer [9]	CA-CFSL [16]	DCFSL [39]	Gia-CFSL [45]	TGP-CFSL
0.1	OA	61.57±3.89	56.47±4.13	72.84±3.61	72.57±2.72	79.00±0.04	56.87±4.44	78.12±4.19	74.92±1.91	77.77±5.66	**81.27±2.32**
	AA	72.57±0.74	68.26±0.87	83.10±1.39	75.50±1.09	81.92±3.77	65.06±2.61	81.17±3.33	73.14±0.35	80.95±2.41	**84.82±2.15**
	kappa	52.57±3.60	47.31±3.61	66.38±3.72	64.95±2.96	72.72±5.20	46.59±4.77	72.06±5.21	65.87±2.24	71.64±6.88	**77.60±4.23**
0.3	OA	58.36±2.74	54.74±2.91	69.40±3.50	69.87±1.80	77.15±0.05	58.84±2.52	72.88±3.95	72.09±3.56	75.40±4.61	**78.35±2.44**
	AA	69.67±1.12	65.99±1.35	78.21±0.69	72.25±0.56	80.73±2.89	65.96±1.63	77.85±2.35	75.24±1.64	75.58±2.32	**79.68±5.12**
	kappa	48.85±2.37	45.30±2.09	61.68±4.00	61.45±1.52	70.80±5.76	48.59±2.52	65.74±4.71	64.33±4.44	72.88±5.49	**78.75±1.22**
0.5	OA	56.75±3.40	52.97±4.25	65.34±4.11	66.33±2.20	76.62±0.05	55.43±5.72	67.04±0.79	69.79±1.60	75.10±5.40	**75.87±2.85**
	AA	67.30±0.91	63.93±1.94	75.06±1.85	68.09±1.20	79.37±1.16	64.92±2.51	70.33±2.45	72.82±2.70	77.07±3.15	**76.26±1.95**
	kappa	46.95±2.85	43.37±3.72	57.10±4.67	56.71±2.15	69.91±6.10	45.35±5.83	57.91±0.95	61.29±1.84	68.22±6.37	**76.52±2.75**



(a) (b) (c)

Fig. 10. Classification accuracies with a different number of labeled samples on four target datasets. (a) IP. (b) UP. (c) Salinas.

C. Alignment Performance of TGP-CFSL

We show a 2-D visualization of the raw sample data for UP, IP, and Salinas using the t-SNE method (as shown in Fig. 8). It can be observed that the class separability of the raw samples is poor due to the fact that different land cover classes may have similar features in the spectral dimension, e.g., class no. 2 (grass) and class no. 6 (bare soil) of the UP dataset.

The primary objective of FSL is to discern the shared characteristics among various metatasks by undergoing numerous scenario-based training processes, such as measuring sample similarity. This facilitates the enhancement of class separability across all three TDs post-FSL training. To assess the efficacy of the DGIP module in addressing domain-shifting issues, we conducted comparative experiments by incorporating or excluding the DGIP module from the proposed method. Analyzing the experimental outcomes revealed that the DGIP module significantly enhances class separability. For instance, in Fig. 8(c), the classification boundary between Bitumen and Bricks appears distinctly clearer compared to Fig. 8(b). This underscores the effectiveness of DGIP in improving category separability within the metric space by aligning target feature relations and similarity properties of feature distributions. Furthermore, within the proposed TGP-CFSL, the incorporation of target category information is augmented through edges under various attention weights, owing to the construction of composite neighborhoods via topology. This augmentation results in smoother target features and more uniform distribution of targets across the domain in the metric space, as depicted in Fig. 8(b), (e), and (i).

D. Analysis of Learning Rate

The learning rate directly affects the convergence state of the model, i.e., it controls the learning progress of the network model and determines whether the network can find and how long it takes to find the global optimal solution. Too large a learning rate leads to erratic learning, and too small a learning rate leads to extremely long training times. To investigate the effect of learning rate on the classification performance of the proposed network, Table VII provides the classification results corresponding to different base learning rates in the UP datasets. The experimental results show that both too large and too small learning rates are detrimental to the convergence of the model, leading to a decrease in the classification accuracy of the model, so we choose a learning rate size of 0.0005.

E. Influence of Number of Labeled Samples

All the above experiments are conducted under the few-shot setting of sampling five labeled samples per class in the TD. In this section, we attempt to build the few-shot dataset for model training and validate the different settings by randomly sampling different numbers (1–5) of labeled samples from the TD. Fig. 10 compares the training accuracy of the four datasets for different methods with different few-shot settings. Consistent with our imagination, as the training data increases, the better the model is trained and the higher the accuracy of the test. Moreover, our proposed CA-CFSL always achieves better results in the setting of the same number of labeled samples per class.

F. Robustness Analysis

It is well known that spectral data are usually subject to various attenuations, noise effects, or variations during the imaging process [46]. In order to evaluate the robustness of our proposed method under noisy conditions, we processed the original hyperspectral dataset with the addition of Gaussian noise. First, a noise matrix with the same shape as the original

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TABLE IX
CLASSIFICATION ACCURACY (%) OF THE PROPOSED TGP-CFSL ADJUSTED FOR λ_1 AND λ_2 PARAMETERS ON THREE EXPERIMENTAL DATA

parameter \ Datasets	UP					IP					Salinas				
λ_1	0.1	0.3	0.5	0.7	0.9	0.1	0.3	0.5	0.7	0.9	0.1	0.3	0.5	0.7	0.9
λ_2	0.9	0.7	0.5	0.3	0.1	0.9	0.7	0.5	0.3	0.1	0.9	0.7	0.5	0.3	0.1
OA	82.50	84.14	**84.27**	83.33	82.83	70.38	73.50	74.61	73.28	**76.13**	88.73	89.78	**92.12**	90.19	89.27

data was randomly generated using a normal distribution with zero mean and standard deviation of ω. Then, the generated noise matrix was added element by element with the original data to obtain the noisified hyperspectral data. In order to evaluate the performance of the proposed algorithm under different noise conditions, we conducted experiments on the UP dataset with different comparison methods by adjusting the value of ω, and by adjusting the standard deviation of the noise, experimental scenarios with different signal-to-noise ratios (SNRs) can be simulated to verify the robustness of the algorithm more comprehensively, as shown in Table VIII. The experimental results show that the proposed TGP-CFSL is able to achieve state-of-the-art performance even under noise interference.

G. Influence of Hyperparameters

The weights of $\mathcal{L}_{\text{fsl}}$ and $\mathcal{L}_{\text{MMD}}$ may affect the classification performance in the total loss function of the model training, and in order to analyze the hyperparameters in this part, we set them to different values and then test the OA index of the TGP-CFSL. Each experiment is repeated ten times. We take the values of λ_1 and λ_2 from $\{0.1, 0.3, 0.5, 0.7, 0.9\}$ and $\{0.9, 0.7, 0.5, 0.3, 0.1\}$, respectively, and the OAs of all experimental datasets on the three embedded feature dimensions are listed in Table IX. From the experimental results, the optimal weight of hyperparameters on the UP and Salinas datasets is 1:1, and the optimal weight of hyperparameters on the IP dataset is 9:1.

VI. CONCLUSION

In this work, we propose the TGP-CFSL framework based on FSL to address the cross-domain few-shot classification problem in HSI classification, designing the TGMF and DGIP modules. Specifically, TGMF is divided into two stages: simple clustering using the super-pixel segmentation method to obtain neighbor nodes and CG-Conv module that combines CNN and GCN to capture both local and nonlocal information, which can effectively capture spatial–spectral features and enhance the domain-invariant features of the target through topological relationships. In addition, the DGIP module aligns domain distribution information by aggregating at both the feature and distribution levels, capturing interdomain similarities and aligning differences between domain distributions. Experimental results on three datasets confirm the superior performance of the proposed approach.

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